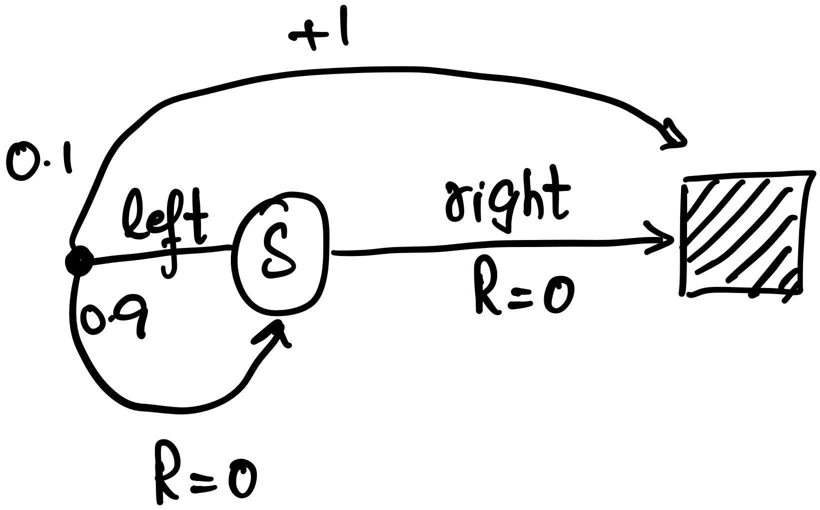

$$\pi(\text{left} \mid s) = 1$$

$$b(\text{left} \mid s) = 1/2.$$

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)}.$$

Episodes:-

	Steps	return	ρ
i:- left $\rightarrow$ terminal	1	+1	$\frac{1}{1/2} = 2$
ii:- left $\rightarrow$ s $\rightarrow$ left $\rightarrow$ t	2	+1	4.
iii. right $\rightarrow$ terminal	1	0	0

iv:- left, left, left, left $\rightarrow$ t 4 1 16.

Ordinary Sampling:-

$\rightarrow$ $V(s) = \frac{1}{n} \sum_i P_i G_i$

$= \frac{1}{3} [2(1) + 4(1) + 16(1)]$

$= 22/3 = \boxed{7.3}$

Weighted:-

$\rightarrow$ $V(s) = \frac{\sum_i P_i G_i}{\sum_i P_i}$

$= \frac{22}{22} = 1$

② Discount aware importance Sampling.

Per step Contribution.

$t=0$ $\quad$ $t=1$.

left $\rightarrow$ terminal

t	γ^t	P_t	$(1-\gamma)\gamma^t P_t$
0	$\gamma^0 = 1$	2	$(\cancel{0.1})(1)(2)$ $= 2$ (2).

~~$0.2 + 0.36 = 0.56.$~~

2

left, left, t

$t=0$ $\quad$ $t=1$.

t	γ^t	$(1-\gamma)\gamma^t G_t$
0	1	$(0.1)(1)(0)$
1	0.9	$+\ 0.9 \times (1)4$

3.6

$0.9 \times 16 \times 1$

$= \underline{11.664.}$